Robot Run Scoring Worksheet

Full 2:30 test runs. Run it like the real match. Write down every run.

A full test run is a practice match: all six missions in order, with the **2 minute 30 second** timer running. Each run gets its own run sheet. After a few runs, the summary pages show our **average, best, worst and middle score**, and which missions fail most often. That tells us what to fix first, and it is proof for the Robot Design judges.

Our mission order

#	Mission	How it scores (rule book)	Max
1	**Mission 9: Research Platform**	Platform raised 10 · camera trap deployed 10 · seed no longer touching the tree 10	**30**
2	**Mission 1: Drone Survey**	Drone no longer touching the mat 20 · *bonus:* LiDAR map completely flipped, scan marker partly in the survey area +10	**30**
3	**Mission 2: Exploding Seeds**	10 for each seed no longer touching the stalk	**10 each**
4	**Mission 15: Biocentric Architecture** (city dock: garden skylight and nesting canopy)	Nesting canopy raised 10 · garden skylight completely in 10 · *city dock bonus* (skylight) +10	**30**
5	**Mission 6: Leafcutter Frenzy**	Ant touching the nest: 10 for each leaf fragment inside the nest	**10 each**
6	**Mission 11: Window to the Past**	Root cover down, touching the mat 20	**20**

Done = the max for that mission. For Seeds and Leafcutter, count the seeds and leaf fragments on the models and write the max here: Seeds _____ Leafcutter _____

Equipment inspection: 20
Robot and all attachments fit in one launch area, under 12 in. tall.

Precision tokens: start with 6
Lose 1 each time we touch the robot outside home. Tokens left → points: 6 or 5 = 50, 4 = 35, 3 = 25, 2 = 15, 1 = 10, 0 = 0

Jobs for each run

1. **Driver** runs the robot and swaps attachments. **Timer** starts the 2:30 clock and calls out the time. **Scorer** fills in the run sheet. Swap jobs every run.
2. **Reset the field** exactly like the start of a match, with every model in its starting spot.
3. **Start the timer and go.** When each mission ends, the timer calls out the time left and the scorer writes it in the **Timer** column.
4. **Stop at 0:00** even if missions are left. Missions not tried count as **Failed**, 0 points.
5. **Score each mission** using the rule book, add up the total, and agree on the **one change** to make before the next run.

✓ Done
Did everything the mission asks and got full points.

½ Partial
Got some points but not all of them.

✗ Failed
0 points: missed, broke, ran out of time or not tried.

Rules that keep the numbers honest

- **Never skip a bad run.** If we only write down good runs, our average lies to us.
- **Change one thing between runs** (code, attachment or how we line up), so we know what made the difference.

Run Sheet Run # _____

Date: _____ **Driver:** _____ **Timer:** _____ **Scorer:** _____

#	Mission	Result	Points	Timer at end	What happened? What should we change?
1	MISSION 9 **Research Platform** Platform up 10 · camera trap 10 · seed off tree 10	☐ ✓ Done ☐ ½ Partial ☐ X Failed	of 30	m : ss left	
2	MISSION 1 **Drone Survey** Drone off mat 20 · bonus: LiDAR map flipped +10	☐ ✓ Done ☐ ½ Partial ☐ X Failed	of 30	m : ss left	
3	MISSION 2 **Exploding Seeds** 10 for each seed off the stalk	☐ ✓ Done ☐ ½ Partial ☐ X Failed	of ___	m : ss left	
4	MISSION 15 **Biocentric Architecture** Canopy up 10 · skylight in 10 · city bonus +10	☐ ✓ Done ☐ ½ Partial ☐ X Failed	of 30	m : ss left	
5	MISSION 6 **Leafcutter Frenzy** Ant at nest: 10 per leaf fragment	☐ ✓ Done ☐ ½ Partial ☐ X Failed	of ___	m : ss left	
6	MISSION 11 **Window to the Past** Root cover down 20	☐ ✓ Done ☐ ½ Partial ☐ X Failed	of 20	m : ss left	

MISSION POINTS	+ INSPECTION	TOKENS LEFT	+ TOKEN POINTS	= TOTAL POINTS
_____ add 1-6	_____ 0 or 20	_____ of 6	_____ max 50	_____ pts

TOTAL TIME USED	✓ DONE	½ PARTIAL	X FAILED
_____ m : ss	_____ of 6	_____ of 6	_____ of 6

Token points: 6 or 5 left = 50, 4 = 35, 3 = 25, 2 = 15, 1 = 10. **Total time used** = 2:30 minus the timer when the robot finished its last mission (2:30 if the timer hit 0:00).

The ONE change for the next run

_____

_____

Who will do it: _____ **Did it help? (next run)** _____

All Runs: Points and Time

Copy the totals from each run sheet into one row. The first row is an example.

Run	Date	Total points	Time used	✓ Done	½ Partial	✗ Failed	Tokens left	The change we made after this run
eg	*11/12*	*145*	*2:22*	*3*	*2*	*1*	*5*	*Slow down the turn before Mission 6*
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								

Our points numbers

Use only the **Total points** column, and don't count the example row.

Average

Add up all the totals, then divide by the number of runs.

_____ pts

Max (best)

The biggest total.

_____ pts

Min (worst)

The smallest total.

_____ pts

Median (middle)

Put the totals in order. The middle one. Even number of runs? Average the two middle ones.

_____ pts

Work space: totals in order, smallest to biggest

What the numbers tell us: is the average close to the max? If not, which missions keep dragging it down?

Why the median? One really bad run (robot fell over) pulls the average down a lot but barely moves the median. If the average is much lower than the median, we have a few disaster runs to look at.

Mission by Mission

For each run, write the mission's result: ✓, ½ or ✗, and its points underneath if there is room. Then count across. The mission with the most ✗ is the one to fix first.

Mission	1	2	3	4	5	6	7	8	9	10	11	12	✓ count	✗ count
1 · Mission 9 Research Platform														
2 · Mission 1 Drone Survey														
3 · Mission 2 Exploding Seeds														
4 · Mission 15 Biocentric Architecture														
5 · Mission 6 Leafcutter Frenzy														
6 · Mission 11 Window to the Past														

What to fix next

#	Mission	Why does it fail or only get partial points?	Fix and who
1			
2			
3			

Time check

Slowest mission (biggest time gap on the run sheets): _____

Average time used: _____ of 2:30

Spare time we could use for another mission (2:30 minus average time used): _____

The time a mission takes is the gap between the timer reading before it and after it. The first mission (Research Platform) starts at 2:30.